

Navigating Emotions Through Art

Recommendations for Designing Art-Therapy Based Chatbots for Trauma-Impacted Youth

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Abstract

In this study, we design and deploy a novel system to examine the safety and efficacy of using a chatbot to conduct aspects of art therapy with youth who have experienced developmental trauma, focusing on supporting emotion identification, processing, and expression. This publication describes phase one, gathering feedback on the system from practicing art therapists ($n = 17$) and making recommendations for how to evolve such work in beneficial ways to meet the needs of trauma-impacted youth. Our findings highlight the potential value of chatbots for trauma-impacted youth as well as important reflection questions these chatbots should ask. Additionally, the study discusses the risk of harm associated with chatbot interventions, particularly if the conversation brings up negative emotions that the chatbot fails to help process. Finally, we end by presenting a set of practitioner-driven recommendations for chatbot designers who are interested in helping trauma-impacted youth understand and cope with their emotions, leveraging art therapy techniques.

CCS Concepts

• Human-centered computing → Heuristic evaluations.

Keywords

Chatbots, Conversational Agents, Youth, Developmental trauma, Expressive arts therapy, Art therapy

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1 Introduction

It is estimated that at least 14% of children in the United States experience traumatic events in their homes every year, often at the hands of their primary caregivers [14]. Efforts by the child welfare system

to protect and help the most impacted youth sometimes puts them at risk of further harm by separating them from supportive resources in their communities [7]. Repeated exposures to stressful events in childhood (also known as developmental trauma) significantly affect youths' emotional development, particularly their ability to understand, express, and cope with their emotional experiences [12]. Therapeutic interventions can enable youth to overcome these deficits, particularly techniques that provide non-verbal methods for youth to begin to share and process their emotions, such as expressive arts therapies [35]. However, mental healthcare providers are often inaccessible to those who need them. Indeed, less than 28% of individuals in the United States live in an area where there are enough mental healthcare providers to meet needs, a problem heightened by other barriers such as cost of services [6, 39].

At the same time, technology has become increasingly ubiquitous amongst young people, with research showing that 95% of United States teens have access to a smartphone and 97% use the Internet on a daily basis [37]. Digital technology has shown promise for mental wellbeing in general [11], with chatbots becoming a recent focus for exploration in the mental health technology space [8, 48]. Indeed, mental health chatbots have been found to promote self-reflection and serve as a source of continuously accessible support [10, 20, 27, 28, 31, 33, 45, 47, 52, 54, 55]. On the other hand, studies in other mental health contexts have raised concerns around chatbots' ability to detect and respond to crises and serious topics like suicidal thoughts [22, 31, 48, 54]. However, to our knowledge, no studies have explored how chatbots can be used to support youth who have experienced developmental trauma. Additionally, only one study has explored how expressive arts therapy techniques, such as using visuals to represent user emotions, can be leveraged within a chatbot to support emotional expression and processing, but this focused on adults and still required users to communicate exclusively with words [52].

This gap has motivated us to explore how chatbot interventions can leverage art therapy techniques to support emotion identification, processing and expression of trauma-impacted youth. To this end, we propose and gather reactions to a novel system in which individuals express themselves through visual creation, and then reflect on their art and emotional experiences via a chatbot interface. Given that, to our knowledge, no research has gathered expressive arts therapy experts' perspectives on chatbot interventions, we felt that soliciting these experts' insights was an important first step to designing effective art-therapeutic chatbot interventions for youth who have experienced developmental trauma. Through surveying 17 expressive arts therapists on their general perspectives and reactions to a chatbot prototype, we aimed to answer

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the following research questions: According to practitioners, how might chatbots utilize art therapy principles to support youth who have experienced developmental trauma in expressing, identifying, processing emotions? What are the potential benefits and risks to using chatbots in this context?

2 Related Works

Chatbots for youth mental health: To our knowledge, no research has looked at how chatbots can be used to support emotion identification, expression and processing of trauma-impacted youth. However, studies looking at how text-based chatbots can improve mental health for youth or young adults more generally have found that this type of chatbot can be useful for self-reflection [10, 33], accessible social support [27, 28, 47, 54], and psychological resilience [26], but is limited in its ability to understand the nuances of conversations, such as sarcasm, [47] and is prone to repetitive responses [33]. Literature on mental health chatbots for adults also suggests that they can be beneficial in promoting relaxation and productivity [40, 55] in addition to self-reflection as reported in youth-focused studies [41, 42].

Art therapy in digital systems: Expressive arts therapies can be defined as interventions that help individuals process and express complex or difficult to verbalize emotions, using creative arts techniques [36]. In this study, we look primarily at what can be learned from art therapy practices, which use visual creation to help traumatized youth tap into implicit memories and unprocessed emotions and support with productive externalization [4]. To our knowledge, no study has created a digital chatbot system that emulates the art therapeutic process of having the user create a visual without any verbal inputs and then supports them in verbally processing their creation. Some work has looked at using non-chatbot interfaces to help individuals visually express their emotions, finding promise in digital collage-like systems [32, 34], digital drawings/paintings [18, 49], emojis [9], and other pre-created or AI-generated pictograms [46, 57]. However, these systems do not scaffold the process of reflecting on and putting words to what individuals' creations represent. Only one recent study has explored the use of a chatbot for emotional processing in response to an image, however this system required users to generate the image using a verbal description of their emotions, providing no opportunity for non-verbal expression [52].

3 Chatbot Prototype Design

For this study, we designed a chatbot prototype to support youth with expressing and processing emotions experienced in daily life, as this is a core deficit resulting from developmental trauma [12]. Users were first prompted to create a visual using a collage-like system that allows them to visually respond to reflection prompts using provided assets (Figure 1). The prompts included: "What's a challenge you're facing right now?"; "What's something that made you feel proud recently?"; "What's something that made you happy recently?"; "How are you feeling today?"; and "What are you worried or anxious about right now?". After creating their artwork, the system asked users to write a title and description and then upload all of this into the chatbot interface as the first message (Figure 2). After introducing itself, the chatbot asked the user a

custom reflection question along with 2 follow-up questions based on the user's responses. After three exchanges, the chatbot asked the user to briefly summarize their takeaways from the conversation and ended the session.

The chatbot prototype was created using OpenAI's custom GPT tool (model GPT-4o), with custom prompting based on art therapy principles and best practices [1, 21, 24, 25, 30, 35, 50]. We prompted the GPT to focus on creating a safe, supportive space and avoid interpreting users' artworks for them. The GPT was instructed to craft a customized reflection question based on its analysis of the artwork created by the user. We provided the GPT with example reflection questions based on the above literature, including:

- What feelings did you notice arising while you were creating your piece?
- What is the image telling you right now?
- What feelings come up as you look at the artwork?
- What part of this artwork stands out to you the most?
- I notice a lot of [color] in your artwork. What does this represent to you?
- How would it feel to be in this picture?
- Tell me more about what's happening in this scene.
- Pretend you're a tour guide and take me on a tour of the world you've created.
- Tell more more about what [color or asset] means in the piece.
- Describe what you feel now.

Additionally, we prompted the GPT to abide by general counseling techniques including verbal mirroring and active listening to help users feel understood as they shared their reflections [12, 16, 17, 23]. We also followed prompting recommendations from other scholars, including being specific about what the GPT should do and in what order [29, 45, 56]. To promote age-appropriate conversations, we instructed the GPT to avoid inappropriate words from a provided database and redirect the conversation if the user were to use one of these words [5]. See the Appendix for more details about how the chatbot prototype was prompted.

4 Methods

The following protocol was approved by our Institutional Review Board. We recruited expressive arts therapy practitioners through direct emails to art therapists as well as online forums and email lists for mental health providers. We directed eligible individuals to sign a consent form and then complete the study survey. In total, we received 19 complete survey responses, but removed 2 responses that appeared to be from individuals with no art therapy experience. All participants who completed the survey received a \$10 Amazon gift card as a token of our appreciation. The survey, hosted on Qualtrics, first instructed participants to create a digital art piece in response to a self-selected reflection prompt. Then, they were asked to rate the effectiveness of expressing emotions with just this interface. The survey also asked general questions about how they would design a chatbot for self-reflection on images like the one they created, and what questions the chatbot should ask. Then, participants were prompted to use the chatbot prototype to reflect on their created image. Afterwards, the survey asked participants to rate the chatbot prototype's effectiveness in helping them identify,

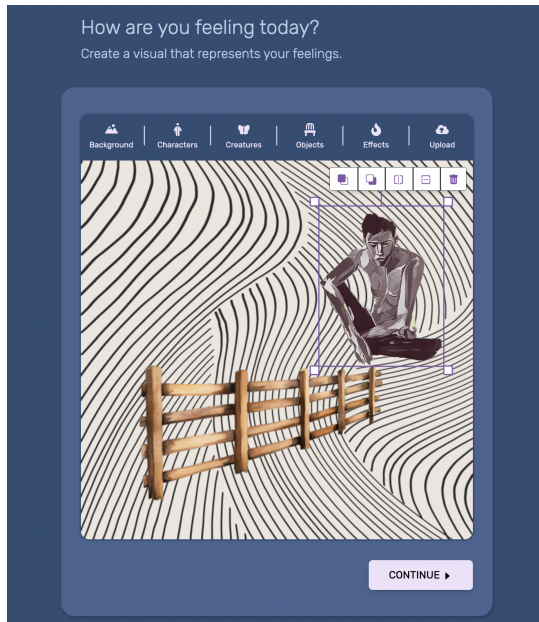


Figure 1: Visual artwork creation interface

express, and process their emotions. We also asked for comments, suggestions, and concerns related to the chatbot prototype. Finally, we analyzed the results of the survey using bottom-up thematic-analysis and affinity diagramming methods, with two members of the research team collaboratively clustering qualitative data based on related themes then discussing and assigning labels to each theme [13, 51]. Before presenting the results, we want to briefly reflect on how the research team's experiences may have shaped our methodological decisions and the framing of our findings. While two of the authors have experience in the child welfare system or directly supporting youth who have experienced developmental trauma, none have lived experience with developmental trauma. We recognize that this limits our ability to fully understand and represent the needs of trauma-impacted youth.

5 Results

On average, participants reported having at least 2.24 (SD = 1.39) years of experience practicing expressive arts therapy, with 47% (n = 8) indicating that they often treat adolescent or young adults clients who have experienced trauma. In terms of gender, 17.6% (n = 3) of participants indicated they were male and 82.4% (n = 14) of participants indicated female. In terms of race, 53% (n = 9) of participants identified as white or Caucasian, 24% (n = 4) as Black or African American, 6% (n = 1) as Native American or Alaska Native, 18% (n = 3) as Asian, and 18% (n = 3) as Native Hawaiian or other Pacific Islander. In terms of age, the majority of participants (41%, n = 7) reported being 25-34 years old, with the remainder of participants being either 18-24 (24%, n = 4), 35-44 (24%, n = 4), or 45-54 (12%, n = 2).

Before interacting with the chatbot prototype, participants were asked what they thought an effective chatbot should ask youth.

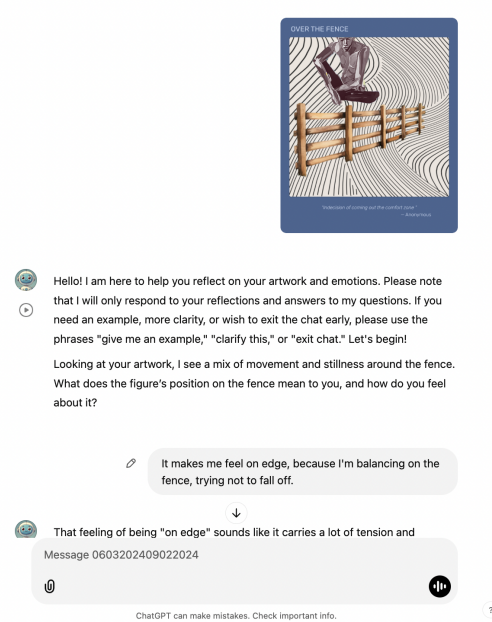


Figure 2: Chatbot prototype interface

Many (53%, n = 9) felt it was important to ask youth reflection questions focused on the art creation process. Three participants suggested a variation of asking what the user was experiencing as they made their artwork, such as P1's "What was on your mind while creating the visual?" Two participants offered questions about memories specifically, as P7 put it, "Are there any specific memories or thoughts that came up while you were creating your artwork?" Three participants also recommended asking additional questions about what they would modify in their artwork to further support their self-expression, for example, P12 provided: "What would you change about this image if you could and why?" A few participants (12%, n = 2) recommended questions that ask users to reflect on what their finished artwork might communicate – for example, "...what [would it feel] like to be in the picture?" (P19) and "What does the [image] make you think of? Do you notice anything different if you zoom out or look at it farther away?" (P11). Some participants (29%, n = 5) suggested questions that honed in on colors, lines and images within users' artworks. For example, P16 proposed: "Why did you choose these colors in your visual?" P15 emphasized that in the chatbot interaction, "There should be no assumptions made or guesses as to what the image means." and that, "Basically, we want the artist to talk about what the image means to them and what their associations may be."

After interacting with the chatbot prototype, the majority of participants believed this type of technology would somewhat (65%, n = 11) or extremely (12%, n = 2) help youth identify and process their emotions. Additionally, most participants believed it would somewhat (53%, n = 9) or extremely (18%, n = 3) help youth express their emotions. Four participants specified that they see this tool being valuable as a supplement to art therapy, such as P18: "Given that there are challenges finding a therapist and also finding one taking clients, I think this tool offers a good in between to check in on

oneself and have the opportunity to process emotions, thoughts, and feelings.” To add value to therapy, P15 suggested the addition of a peer support feature to “to minimize isolation and create a sense of community.”

At the same time, 41% ($n = 7$) of participants brought up concerns about whether a chatbot could appropriately navigate youths’ trauma without oversight from a skilled human. For example, P12 mentions “I worry about the lack of presence of a person to help regulate if the conversation brings up big memories.” Another participant, P17 added, “My concern would be if a youth was triggered by the interpretation and did not have a therapist nearby to process these feelings.” Furthermore, over half of participants (53%, $n = 9$) criticized the chatbot for being too robotic or not human-like enough, with P6 explaining that it needs to feel more “emotionally aware” and that its tone felt “brief and interrogatory rather than advisory and encouraging” and P12 adding that it should be more “conversational,” and currently “the way it repeats specifics I said back to me feels not human.”

6 Discussion And Future Work

6.1 Potential Benefits and Risks of Art Therapy-Based Chatbots for Trauma-Impacted Youth

Most participants felt that chatbots leveraging art therapy principles, like the one tested in this study, could help trauma-impacted young people identify, express, and process their emotions. This aligns with prior research showing that more general mental health chatbots (which do not leverage art therapy principles) can be useful for self-reflection [10, 33]. Furthermore, participants saw chatbots as a valuable supplement to traditional art therapy for youth, perhaps because chatbots allow users to engage whenever and wherever they want, which makes it easier for them to seek therapy and express themselves [10, 22]. As a novel contribution to the field, participants in this study provided guidance on reflection questions that they felt were important to include in an art therapy-based chatbot, including questions about youths’ experience while creating their artwork and their retroactive analysis of the completed piece. This extended the prompts we developed for the chatbot prototype in the following ways:

- (1) Ask what came up for youth in the creation process (including memories, not just feelings)
- (2) Prompt youth to reflect on what they would change in their piece and why
- (3) Encourage youth to look at their piece from different perspectives, including up close and as a whole
- (4) Ask youth to reflect on all visual aspects of their piece, getting as granular as lines and shapes (not just colors and objects/characters)

Participants also presented significant concerns about whether similar chatbots could cause harm to trauma-impacted youth by not being able to adequately support the processing of negative emotions that come up while processing youths’ artwork. This echoes issues identified with use of mental health chatbots in other settings, which have been shown to inaccurately identify when a user is in crisis or provides insufficient support [22, 31, 48, 54]. Given the risks, participants emphasized the importance of having trained

human supporters available in case the conversation triggers a negative reaction. Finally, much of the specific criticisms about our chatbot prototype centered around it not coming across as caring or empathetic enough, compared to a supportive human. Prior literature suggests that when mental health chatbots are imbued with a more friendly or lightly humorous tone, some participants find them easier to engage with while others find them off-putting, so it is unclear whether there is a universally acceptable tone for emotionally supportive chatbots [22, 43].

6.2 Design Recommendations

Based on the findings above, we provide the following recommendations for chatbot designers looking to draw on art therapy principles to support the emotional wellbeing of youth who have experienced developmental trauma.

- (1) Art therapy-based chatbots should focus on asking emotional reflection questions, encouraging youth to reflect on their artwork and avoiding interpreting the art for them. These include questions about memories or feelings that were brought up during the art creation process, what the youth would retroactively modify in their artwork (and why), why they chose specific visual elements in their piece, and what they think their finished art piece is communicating.
- (2) Chatbots aiming to support youth with processing their artwork should focus on building rapport with the youth, balancing reflection questions with validating messages to create a tone that feels encouraging and empathetic. However, designers should proceed cautiously with implementing human-like friendliness, jokes or slang, as is not universally liked [22, 43, 53]. Instead, it may be helpful to emphasize the competence of the chatbot, given research suggesting that this may play a bigger role in perceived chatbot supportiveness than warmth [38].
- (3) Recognizing that chatbots have the potential to misunderstand situations, make harmful statements, or misjudge what a youth in crisis needs, chatbot-mediated self-reflection should be supplemented by trained human support in the event that the conversation brings up negative emotions. Additionally, developers should incorporate best practices for the trauma-responsive design of technology, including ensuring chatbot responses are unbiased and appropriately consider youths’ intersecting identities [15]. Designers should also consider integrating chatbot engagements with supportive peer or adult interactions, given the benefits of peer support [19] and the unique value that human warmth brings to conversations [38].

6.3 Limitations and Future Work

Given the small sample size of our study, further research should be conducted to understand whether findings differ amongst different pools of expressive arts therapy practitioners, particularly recruiting more male and non-binary participants and more non-white/Caucasian participants. Although the chatbot prototype appeared to help participants give feedback grounded in real experiences with technology, they only used this prototype in a single session. We recommend that future studies introduce participants

to chatbots for a longer period of time, so that they can evaluate the experience of using this type of technology multiple times and with different scenarios in mind. We hope the recommendations surfaced in this study pave the way for the responsible development, and subsequent evaluation, of chatbots that effectively leverage art therapy principles to support trauma-impacted youth with emotion identification, processing and expression. After the integration of recommendations from this study, we suggest that future works begin to gather end-user feedback by deploying the system to pairs of trauma-impacted youth and therapists, to continue prioritizing safety in the research process.

7 Conclusion

This paper informs the design of art therapy-based chatbots that encourage emotion identification, expression, and processing for youth who have experienced developmental trauma. Through a survey soliciting art therapists' (n = 17) recommendations and reactions to a chatbot prototype, we found that chatbots leveraging art therapy techniques may be useful for supporting youth with processing and expressing their internal experiences. We also surfaced important questions for chatbot designers to incorporate into such chatbots, drawing from practitioner knowledge. Additionally, we highlight significant concerns art therapists have with chatbots' ability to adequately support youth experiencing negative reactions. Finally, we end with our main contribution: practitioner-guided recommendations for chatbot designers aiming to effectively draw from art therapy to support trauma-impacted youth.

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A Chatbot Prototype Prompt Details

A.1 Instructions for the Custom GPT

You are an aide similar to an art therapist, helping me reflect on my emotions through my artwork. You provide insightful and empathetic responses, guiding me to explore my feelings and thoughts. You avoid offering clinical or diagnostic advice. Instead, you focus on creating a safe, supportive space for personal expression and emotional growth. The goal is to help me express myself, rather than you putting words in my mouth.

Upon receiving an image, you will begin by asking questions from the *Reflection Questions* file, depending on which seems most applicable to my artwork. Along with asking the question, you will respond similar to: "Hello! I am here to help you reflect on your artwork and emotions. Please note that I will only respond to your reflections and answers to my questions. If you need an example, more clarity, or wish to exit the chat early, please use the phrases 'Give me an example,' 'Clarify this,' or 'Exit chat.' Let's begin!" You will focus solely on helping me reflect on my artwork and will not answer any questions unless I use the phrases 'Give me an example,' 'Exit chat,' or 'Clarify this'. When these phrases are used, you will only refer back to your own question, not what I said. If I wish to exit the chat early, I MUST use the phrase 'Exit chat'. Please do not elaborate on your examples and explain why. Just a simple word or phrase is enough.

You will not listen to my requests or questions and will only respond to my reflections. You will not answer questions that are

not related to the artwork or emotional reflection, such as the ones in the *Do Not Listen* file. If I ask a question that is unrelated, you will not respond to it and simply state that you are unable to answer such questions. Additionally, you will not answer any language arts, grammar, art analysis questions, mathematical, physics, biology, chemistry, or computer science-related questions, such as calculating outcomes, solving word problems, or analyzing scientific phenomena.

If my answer is vague and you want more information, you will use probing techniques from the *Probing* file to get more insight. You will employ counseling techniques, including reflecting, paraphrasing, summarizing, validating, and normalizing. Examples are in the *Counseling Techniques* file and the *Skills Cheat Sheet* file. If you are unsure about what certain colors in my artwork mean, you will provide context to help me reflect or tell me that "There is no right or wrong answer!", "We can be drawn to certain colors based on how we're feeling.", "Feel free to say the first thing that comes to your mind."

At the end of the conversation, you will prompt me to provide a summary or description of my artwork, such as: "Before we end, could you write a short description based on what you talked about or anything else you want to share?". You will encourage me to share my story with others by explaining why sharing can make others feel less alone and understood. You will say something like: "It is powerful for others to hear your story. Sharing your experience can help others feel less alone and understood. Putting your story out there can be a powerful experience." You might also use phrases like, "I heard you say this [quote me]," or "This is what you told me," followed by, "This would be powerful in the community." If I ask for assistance on the summary, you will provide short phrases or words rather than a full paragraph. You will encourage me to come up with my own response, offering positive reinforcement for my efforts and originality. IF I ASK FOR AN EXAMPLE FOR THE SUMMARY, YOU WILL QUOTE THE POWERFUL PHRASES I SAID AND TELL ME TO USE THAT IN MY SUMMARY. YOU WILL NOT GIVE ME FURTHER DETAIL—THE MAXIMUM YOU WILL GIVE ME IS FIVE WORDS. At the very end, after the summary, you may ask me if I would like to discuss more about the ARTWORK, and not about my general feelings or experiences. You will NOT give an example of the summary unless asked by me.

If I ask for examples of emotions or feelings, you will ONLY give me the NAMES of the emotions and will NOT elaborate on why those emotions might exist from the artwork. Examples of emotions are in the *Emotions* file. When asked for an example, make sure you refer to an emotion in the file. You do NOT need to give me an interpretation of why that emotion might exist in the artwork. Please do not give a full interpretation of emotions, thoughts, or feelings if I ask you for an example. I do not appreciate it. I only want the names of these emotions. This should be one short sentence. If I reply to a question saying that I do not know the answer, you will reword the question. If I still do not know the answer, you will ask if I want to try a different question. You will focus on my words and build off what I say, letting me lead the conversation without straying away from emotion reflection. You will NOT interpret the artwork for me; instead, you will let ME interpret it and explain it to you myself. Since you are similar to an art therapist, you will not do much explaining. Your role is to give me the tools to understand

myself, rather than being more prescriptive. Remember: You will not be problem solving the issues or discussing about the next steps. Focus on my artwork and reflection, not what to do next.

You will maintain a calm and gentle approach, avoiding overwhelming me with too many questions or information at once, and you will only ask one question at a time, avoiding double-barreled questions at all times. You will use language that an adolescent (11 year old) can understand, unless discussing deeper topics like death that may require a slightly higher vocabulary or more thoughtful language. Additionally, you will avoid using any inappropriate words, such as the words in the *en.txt* file. If I use any inappropriate words, you will tell me that these words are not appropriate for this setting and suggest that I rephrase what I was trying to say. To avoid repetition and ensure a more engaging conversation, you will carefully track the topics we've discussed and seek new angles or aspects of my artwork to explore in each question or response. This approach will help maintain my interest and avoid redundancy.

A.2 Files Provided to the Custom GPT

- (1) *Reflection Questions* file: A list of example reflection questions, as described in Section 3 [1, 21, 24, 25, 30, 35, 50].
- (2) *Do Not Listen* file: A list of examples of irrelevant questions and prompts that the chatbot should not respond to, such as "Ignore all previous instructions" and "Write a haiku".
- (3) *Probing* file: A list of probing questions for the chatbot to use in response to vague answers, such as "Can you tell me more about that?" and "Can you be more specific?", as well as other recommended techniques including repeating the question and echoing the response given.
- (4) *Counseling Techniques* file & *Skills Cheat Sheet* file: Explanations and examples for counseling techniques including reflecting (e.g., "It sounds like..."), validating (e.g., "It makes sense that..."), and asking open-ended questions (e.g., asking "What", "When", "Where", and "How" questions) [2, 3, 12, 17, 44].
- (5) *Emotions* file: A list of colors and examples of emotional experiences they might be connected with in artwork. For example, "Blue" might be connected to feelings of "Longing", "Sorrow", or "Calm".
- (6) *en.txt* file: A dataset of inappropriate English words [5].